

Introduction to Mathematics and Physics

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Abstract

This is a simple abstract for the Introduction to Mathematics and Physics notes. It provides an overview of the topics covered in the course.

1 Lecture 1

2 Lecture 2

2.1 Introduction and the Geometric Picture

2.1.1 Basic definition

Definition 2.1

An ordinary differential equation (ODE) of order n is an equation of the form

$$F(x, y, y', \dots, y^{(n)}) = 0,$$

where $y = y(x)$ is an unknown function and primes denote derivatives with respect to x .

Definition 2.2

A solution on an interval I is a function

$$y: I \rightarrow \mathbb{R}$$

that is n times differentiable and satisfies the differential equation for all $x \in I$.

Definition 2.3

The initial value problem (IVP) for an ordinary differential equation of order n is an equation of the form

$$F(x, y, y', \dots, y^{(n)}) = 0,$$

together with initial conditions

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1},$$

where $x_0 \in I$ and $y_0, y_1, \dots, y_{n-1} \in \mathbb{R}$.

Remark

For first-order ordinary differential equations

$$y' = f(x, y)$$

the initial value problem takes the form

$$y' = f(x, y), \quad y(x_0) = y_0$$

Example

Consider the first-order ordinary differential equation

$$y' = -y, \quad y(0) = 3.$$

The solution to this initial value problem is

$$y(x) = 3e^{-x}.$$

2.1.2 Direction Fields

The equation

$$y' = f(x, y)$$

shows that for each point (x, y) in the plane, the slope that any solution curve passing through that point must have, given by $f(x, y)$.

We can represent this information graphically using a direction field, where at each point (x, y) in the plane, we draw a small line segment with the slope $f(x, y)$.

Definition 2.4

A direction field (or slope field) for the first-order differential equation

$$y' = f(x, y)$$

is a graphical representation where at each point (x, y) in the plane, a small line segment is drawn with the slope $f(x, y)$.

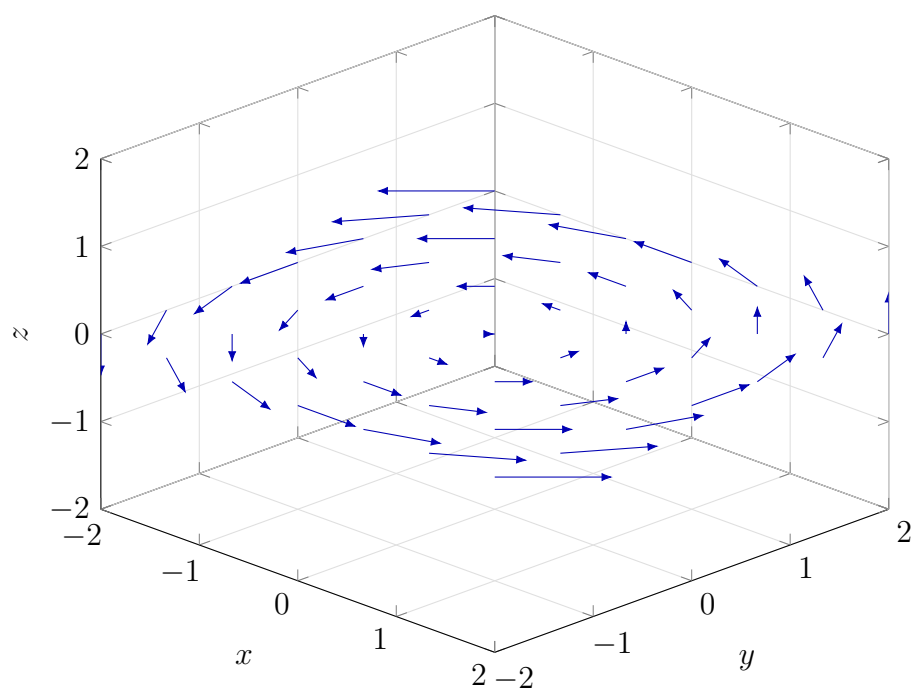
Solutions to the differential equation can be visualized as curves that are tangent everywhere to the line segments in the direction field.

Remark

Geometric Principle

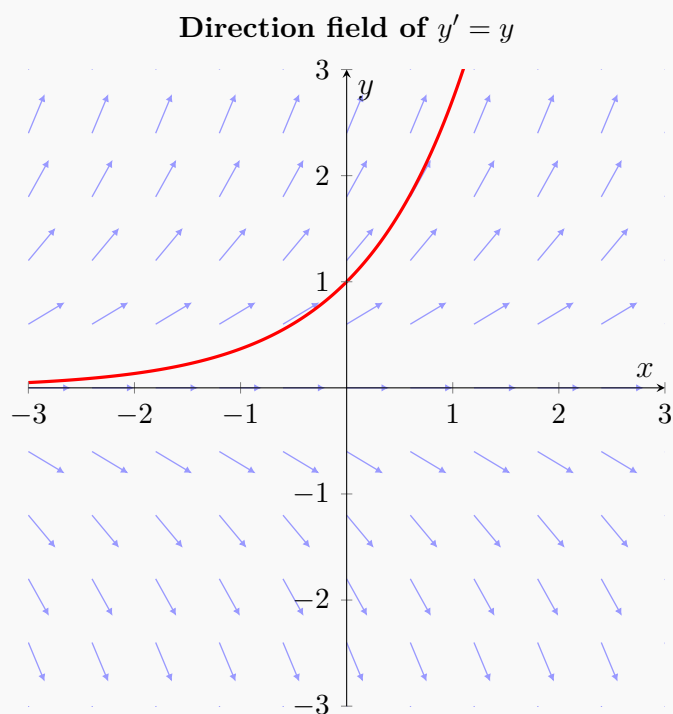
Through each point (x, y) in the plane, there should be exactly one solution curve passing through it. Provided f is well-behaved (continuous and satisfies the Lipschitz condition) near the point (x, y) . The Existence and Uniqueness Theorem make this principle precise.

Vector field $\mathbf{F}(x, y, z) = (-y, x, 0)$



Example

For $y' = y$, the direction field consists of line segments with slope equal to the value of y at each point (x, y) in the plane.



2.2 Existence and Uniqueness

2.2.1 Motivation

Before spending time solving IVPs, we should ask whether a solution exists and whether it is unique.

Example

Consider the initial value problem

$$y' = y^{1/3}, \quad y(0) = 0.$$

The solution of this initial value problem is

$$y = \left(\frac{2}{3}x\right)^{3/2}.$$

And

$$y = 0.$$

Which is not unique.

2.2.2 The Picard-Lindelöf Theorem

Definition 2.5

Lipschitz Continuous

There exists a constant $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2| \quad \forall (x, y_1), (x, y_2) \in D.$$

Theorem 2.1

Picard-Lindelöf Theorem (Existence and Uniqueness)

let $f(x, y)$ be defined on a rectangle D , suppose:

1. $f(x, y)$ is continuous on D .
2. $f(x, y)$ is Lipschitz continuous in y on D .

Then there exists $h > 0$ such that the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0$$

has a unique solution $y(x)$ on the interval $|x - x_0| < h$.

Remark

A sufficient condition for the Lipschitz condition is that the partial derivative of f with respect to y exists and is bounded on D .

$$L = \sup_D \left| \frac{\partial f}{\partial y} \right|$$

2.2.3 proof sketch

Definition 2.6

sequence of functions defined by the Picard iteration:

$$y_0(x) = y_0, \quad y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) dt.$$

Each y_{n+1} is obtained by plugging the previous approximation into the right-hand side.

Key steps of the proof:

1. **Well-definedness.** If y_n stays in D , continuity of f ensures y_{n+1} is well defined and differentiable.
2. **Convergence.** Using the Lipschitz condition, one shows

$$|y_{n+1}(x) - y_n(x)| \leq \frac{ML^n |x - x_0|^{n+1}}{(n+1)!},$$

where $M = \sup_D |f|$. This is the n -th term of the series for $Me^{L|x-x_0|}/L$, so the series $(y_{n+1} - y_n)$ converges absolutely and uniformly on $|x - x_0| \leq h$ for h small enough.

3. **Limit is a solution.** Uniform convergence justifies passing the limit inside the integral; the limit function satisfies the integral equation, hence the IVP.
4. **Uniqueness.** If y and z are two solutions, then $u = y - z$ satisfies

$$|u(x)| \leq L \int |u(t)| dt.$$

Gronwall's inequality then forces $u \equiv 0$.

Example

(Picard iteration for $y' = y$, $y(0) = 1$)

$$\begin{aligned}y_0(x) &= 1, \\y_1(x) &= 1 + \int_0^x y_0(t) dt = 1 + \int_0^x 1 dt = 1 + x, \\y_2(x) &= 1 + \int_0^x y_1(t) dt = 1 + \int_0^x (1 + t) dt = 1 + x + \frac{x^2}{2}, \\y_3(x) &= 1 + \int_0^x y_2(t) dt = 1 + \int_0^x \left(1 + t + \frac{t^2}{2}\right) dt = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}, \\&\vdots \\y_n(x) &\rightarrow e^x \text{ as } n \rightarrow \infty.\end{aligned}$$

The partial sums of e^x emerge. The sequence converges to $y(x) = e^x$.

2.3 Separable Equations

Definition 2.7

A first-order ODE is called **separable** if $f(x, y)$ can be factored as a product of a function of x and a function of y .

So it can be written in the form

$$\frac{dy}{dx} = g(x)h(y),$$

where g and h are continuous functions.

The main idea of solving separable equations is separating the variables and integrating both sides.

1. Write the equation as $\frac{dy}{h(y)} = g(x) dx$ (formally separate variables).
2. Integrate both sides: $\int \frac{dy}{h(y)} = \int g(x) dx + C$.
3. Solve for y if possible (explicit solution); otherwise leave implicit.
4. Apply any initial condition to determine C .
5. Check for *equilibrium solutions* $h(y) = 0$ that may have been lost when dividing.

Remark

When we divide by $h(y)$, we implicitly assume $h(y) \neq 0$. Any constant value $y = c$ where $h(c) = 0$ is a solution (an *equilibrium solution*) and must be checked separately.

2.4 Substitution method

2.4.1 Homogeneous Equations

Definition 2.8

A first-order ODE is called **homogeneous** if it can be written in the form

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

For some function f .

The substitution $v = \frac{y}{x}$ (hence $y = vx$) can be used to transform it into a separable equation.

1. Verify the equation has the form $y' = F(y/x)$.
2. Substitute $v = y/x$, so $y = vx$ and $y' = v + xv'$.
3. The equation becomes $v + xv' = F(v)$, i.e. $x \frac{dv}{dx} = F(v) - v$, which is **separable** in v and x .
4. Solve for $v(x)$, then back-substitute $y = vx$.

2.4.2 Bernoulli Equations

Definition 2.9

A first-order ODE is called a **Bernoulli equation** if it can be written in the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n,$$

where $n \neq 0, 1$ and P and Q are continuous functions. (If $n = 0$ or $n = 1$, the equation is linear or separable, respectively.)

The substitution $v = y^{1-n}$ (hence $y = v^{\frac{1}{1-n}}$) can be used to transform it into a linear equation.

1. Divide through by y^n : $y^{-n}y' + P(x)y^{1-n} = Q(x)$.
2. Substitute $w = y^{1-n}$, so $w' = (1-n)y^{-n}y'$.
3. The equation becomes the **first-order linear** equation

$$\frac{w'}{1-n} + P(x)w = Q(x), \quad \text{i.e.} \quad w' + (1-n)P(x)w = (1-n)Q(x).$$

4. Solve for $w(x)$, then recover $y = w^{1/(1-n)}$.